Xavier Werth

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Objective

Motivated Computer Engineering student at Georgia Tech seeking a Summer 2026 Robotics and Autonomous Systems Internship, with hands-on experience in embedded systems, computer vision, and software development. Eager to apply interdisciplinary skills in perception, controls, and data-driven analysis to build reliable, real-world autonomous technologies while contributing to a fast-paced engineering team.

Education

Georgia Institute of Technology | Atlanta, GA *August 2025– Present*
Bachelor of Science in Computer Engineering Expected Graduation, May 2029
Gold Scholar (Awarded to top 2% of incoming students)

Skills

Programming: Java, Python, C++, JavaScript, HTML, CSS, SQL, Kotlin
Back-end/Tools: PyTorch, NumPy, Unity, Git, React, Flask, Springboot, MATLab, CSVis, ROS

Experience

RoboNav | Atlanta, GA *August 2025 - Present*
Electrical Subteam
A student organization focused on designing and programming autonomous navigation systems for ground and aerial robots.

- Engineered end-to-end electromechanical subsystems, from initial component selection and circuit design to the seamless integration of hardware and mechanical assemblies.
- Developed embedded firmware to program microcontroller logic, process sensor data, and implement precise control systems for various motors (DC, servo, stepper).

Lighter Than Air Autonomy | Atlanta, GA *August 2025 - Present*
Computer Vision Subteam
Research and development lighter-than-air aerial vehicles (blimps), focusing on navigation and indoor flight autonomy.

- Implemented computer vision pipelines in Python using the CSVis library to process, detect, and track visual features for autonomous lighter-than-air vehicles.
- Analyzed camera data to improve localization, obstacle awareness, and system robustness, collaborating with autonomy controls teams to integrate vision outputs into real-time decision-making.

Projects

MyWatkinsville *May 2025*
Developed MyWatkinsville, a geolocation-driven mobile app using Jetpack Compose, JavaScript, and Kotlin to enhance cultural engagement in Watkinsville, GA.

- Integrated the Google Maps API for core location services and utilized a Firebase backend to manage and sync real-time event and business data.
- Achieved over 500 downloads and was officially adopted by the local tourism department, earning a second-place finish in the Congressional App Challenge.

Research

EV Delivery Fleet Feasibility Study *May 2025*
Conducted a comprehensive feasibility study to assess the economic and logistical viability of transitioning commercial delivery fleets to electric vehicles (EVs).

- Leveraged PyTorch and NumPy to process, clean, and analyze large datasets, performing numerical simulations to model vehicle range, operational costs, and charging requirements.
- Created compelling data visualizations using MATLab and Seaborn to illustrate model outputs and key financial metrics within the final feasibility report.

Relevant Coursework

Digital System Design, Data Structures and Algorithms, Linear-Algebra, Object-Oriented Programming, Physics I + II